

2MVA Problem Sheet 2
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1. The domain T is the solid tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(1, 1, 0)$ and $(1, 0, 1)$. Sketch the domain T and evaluate the following triple integral:

$$I = \iiint_T xyz dx dy dz .$$

[20]

2. Change to polar coordinates and compute the integral

$$I = \int_0^1 dx \int_{\sqrt{3}x}^{\sqrt{4-x^2}} x^2 y (x^2 + y^2) dy$$

[20]

3. Consider the following triple integral

$$I = \iiint_T z^3 dx dy dz ,$$

where T is the domain bounded by the planes $x=0$, $y=0$, $z=0$ and $x+y+z = 2$.

- (a) Express the integral as repeated integrals using Cartesian coordinates.
- (b) Express the integral as repeated integrals using cylindrical coordinates.
- (c) Express the integral as repeated integrals using spherical coordinates.
- (d) Calculate the triple integral using one of the three coordinate systems.

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4. Consider the following triple integral

$$I = \iiint_S z dx dy dz ,$$

where the domain S lies in the first octant (i.e., $x, y, z \geq 0$) between the two surfaces $z = x^2 + y^2$ and $x^2 + y^2 + z^2 = 2$.

- (a) Express the integral as repeated integrals using Cartesian coordinates.
- (b) Express the integral as repeated integrals using cylindrical coordinates.
- (c) Express the integral as repeated integrals using spherical coordinates.
- (d) Calculate the triple integral using one of the three coordinate systems.
- (f) Using the result of (d), or otherwise, calculate the following integral

$$J = \iiint_T (z + x \sin(y^2)) dx dy dz ,$$

where the domain T lies between the two surfaces $z = x^2 + y^2$ and $x^2 + y^2 + z^2 = 2$.

[35]